



# PRAMAY | BHATPAHRI

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## Research Profile

Early-career researcher in computational granular mechanics with a doctoral focus on Discrete Element Method (DEM) simulation of dense granular assemblies and DEM-calibrated constitutive modelling of small-strain elastic moduli. Research integrates triaxial DEM with Hertz–Mindlin and Cundall–Strack contact models, audited elastic-window extraction of  $E$ ,  $G$ ,  $K$ , and physics-aware regression to produce separable, support-limited stiffness laws across confining stress, void ratio, and particle-modulus regimes. Current publication record includes two published journal articles, two published conference papers, one journal manuscript under review, and one manuscript in preparation. Empirical constitutive modelling is combined with neural-network and MLP surrogate models supported by AIC/BIC selection and grouped cross-validation; interests span geomechanics, wave propagation in granular media, shock absorption, and computational materials modelling.

## Education

2020 – Present | Indian Institute of Science, Bengaluru  
**Ph.D., Aerospace Engineering**

- › Advisor: Prof. S. Gopalakrishnan | Doctoral research in computational granular mechanics and DEM-calibrated constitutive modelling.
- › Focus: DEM-based constitutive modelling, small-strain stiffness extraction from triaxial simulations, contact-law-dependent fabric and force-network response, physics-aware and data-driven modelling of elastic moduli.

2014 – 2016 | National Institute of Technology, Raipur  
**M.Tech., Machine Design (Honours)**

- › Thesis: *Fatigue and Fracture Behavior of API X65 Line Pipe Steel*; collaboration with R&D Division, Tata Steel, Jamshedpur.
- › Methods/instruments: grain-size analysis (Leica), microhardness (Leco), tensile and fatigue testing (Instron), J-integral fracture toughness, Charpy impact, fractography (Olympus BX41M, Leica M165 C), and Abaqus.

2009 – 2013 | CCEM Raipur, CSVTU Bhilai  
**B.E., Mechanical Engineering (First Class with Honours)**

- › Foundation: machine design, mechanics of solids, fluid mechanics, thermodynamics, manufacturing science, turbomachinery, CAD, and finite-element-based structural analysis.
- › Branch topper, B.E. Mechanical Engineering.

## Doctoral Research / Thesis

Indian Institute of Science, Bengaluru | Advisor: Prof. S. Gopalakrishnan  
**Doctoral research in DEM-calibrated constitutive modelling of granular materials**

- › DEM-based constitutive modelling of dense granular assemblies under isotropic confinement and deviatoric loading using YADE.
- › Comparative analysis of Hertz–Mindlin and Cundall–Strack contact models for small-strain elasticity, coordination state, and micromechanical descriptors.
- › Development of separable, support-limited stiffness maps of the form  $Y = k_Y (E_p/E_{p0})^{\alpha_Y} (\sigma_{\text{iso}}/\sigma_0)^{n_Y} F_i(e_0; \theta_{i,Y})$  for DEM-derived elastic moduli.
- › Integration of physics-aware regression with neural-network and MLP surrogate models, supported by AIC/BIC model selection and grouped cross-validation.

- Extension to breakage-aware ensembles and to fabric / force-anisotropy descriptors connecting macroscopic stiffness to microstructural state.

## Research Interests

- **Mechanics:** Granular materials & soil micromechanics, contact mechanics, constitutive modelling, small-strain elasticity
- **Methods:** Discrete Element Method, fabric tensor / force-network anisotropy, particle breakage, wave propagation in granular media
- **Modelling:** Scientific machine learning, physics-aware regression, DEM-to-continuum surrogate modelling, graph-based / topological descriptors

## Publications

### A. Published Journal and Conference Papers

1. **B. Pramay**, Anay Mohan Shembekar, and S. Gopalakrishnan, “Early-loading tangent modulus maps for dense spherical DEM assemblies across contact laws and particle-modulus regimes,” *Particuology*, 2026. Available online 3 October 2026. DOI: [10.1016/j.partic.2026.09.014](https://doi.org/10.1016/j.partic.2026.09.014). [Article](#).
  - Calibrates separable stiffness maps  $Y(\sigma_{\text{iso}}, e_0) = k_Y(\sigma_{\text{iso}}/\sigma_0)^{n_Y} F_i(e_0; \theta_{i,Y})$ ,  $Y \in \{E, G\}$ , across CS/HM contact laws and low/high particle-modulus regimes.
  - Uses grouped cross-validation, AIC/BIC ranking, and physics-aware admissibility checks: positivity, monotonic stiffening with  $\sigma_{\text{iso}}$ , monotonic softening with  $e_0$ , and admissible  $\nu$ .
2. **Pramay B.** and S. Gopalakrishnan, “Separable surrogates for DEM-derived elastic moduli of Hertz–Mindlin and Cundall–Strack granular packings,” *Computers and Geotechnics*, vol. 200, Article 108453, 2026. DOI: [10.1016/j.compgeo.2026.108453](https://doi.org/10.1016/j.compgeo.2026.108453). [Article](#).
  - Published journal article developing a three-variable separable interpolation surrogate for  $E$ ,  $G$ , and  $K$  using particle Young’s modulus  $E_p$ , isotropic confining stress magnitude  $\sigma_{\text{iso}}$ , and initial void ratio  $e_0$ .
  - Reports contact-law-dependent confinement exponents, BIC-based void-ratio-form selection, and design-level leave-one-design-out transfer diagnostics.
3. **Pramay Bhatpahari** and Srinivasan Gopalakrishnan, “Prediction of Small-Strain Properties of Dry Sand Using Curve Fitting and Machine Learning Models,” *EPJ Web of Conferences*, vol. 340, Article 09013, 2025. DOI: [10.1051/epjconf/202534009013](https://doi.org/10.1051/epjconf/202534009013). [Article](#) | [PDF](#).
  - Published conference paper from Powders & Grains 2025; YADE DEM triaxial simulations using the Hertz–Mindlin contact model were used to extract small-strain stiffness and Poisson’s ratio.
  - Compared empirical curve-fitting laws with neural-network models for predicting small-strain moduli as functions of isotropic confining stress and initial void ratio.
4. **Bhatpahari Pramay**, S. Gopalakrishnan, and Anay Mohan Shembekar, “Constitutive Modelling of Granular Materials Using Cundall–Strack and Hertz–Mindlin Contact Models,” *Proceedings of the ASME 2025 Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM2025)*, Houston, TX, USA, May 5–7, 2025. Paper SSDM2025-152166 / V001T03A016. DOI: [10.1115/SSDM2025-152166](https://doi.org/10.1115/SSDM2025-152166). [Article](#) | [PDF](#).
  - Published conference paper comparing Cundall–Strack and Hertz–Mindlin contact models in YADE DEM triaxial simulations.
  - Developed empirical constitutive laws for  $E$ ,  $G$ , and  $\nu$  as functions of isotropic confining stress and initial void ratio.

### B. Manuscripts Under Review

5. **B. Pramay** and S. Gopalakrishnan, “Breakage-Induced Shifts in Pressure Exponents, Void-Ratio Factors, and Separability of Small-Strain Elastic Moduli in Granular Materials,” *Springer Nature journal*, under review.
  - Analyses no-breakage and particle-breakage DEM ensembles with Hertz–Mindlin contacts; fragment stiffness is represented through the dimensionless ratio  $\alpha_E$ .
  - Fits separable laws for  $E$  and  $G$  using trust-region nonlinear least squares, information criteria, grouped K-fold validation, weak-coupling separability diagnostics, and MLP surrogate modelling.

## C. Manuscripts in Preparation

6. **B. Pramay** et al., “Contact-Law-Dependent Coherence of Small-Strain Fabric and Force Anisotropies in DEM Sphere Packings: Descriptor-Family Partitioning Distinguishes Hertz–Mindlin from Cundall–Strack Packings,” manuscript in preparation.
  - Computes 36 anisotropy invariants grouped into contact-fabric, normal-force, tangential-force, force-support, and cross-tensor descriptor families across HM and CS DEM sphere-packing samples.
  - Uses piecewise-linear model fitting, BIC selection, bootstrap stability filtering, label-permutation testing, and leave-one-family-out robustness audits.

## Research Experience

2020 – Present | Dept. of Aerospace Engineering, IISc Bengaluru

**PhD Researcher — Computational Wave Mechanics / Granular Mechanics**

- Developed DEM-based computational workflows in YADE to simulate triaxial compression of dense granular assemblies under controlled isotropic confinement and deviatoric loading.
- Implemented and compared Hertz–Mindlin and Cundall–Strack contact models to quantify how contact physics affects small-strain stiffness, Poisson’s ratio, coordination state, and micromechanical descriptors.
- Extracted early-loading tangent moduli  $E$ ,  $G$ , and  $K$  from stress–strain data using audited elastic-window detection and elastic-consistency checks.
- Built separable, DEM-calibrated constitutive and surrogate models for elastic moduli with explicit dependence on particle Young’s modulus, confining stress, and void ratio.
- Applied nonlinear least squares, trust-region optimization, grouped K-fold cross-validation, AIC/BIC selection, and physics-based admissibility filters.
- Developed neural-network and multi-layer perceptron (MLP) surrogates to compare empirical interpretability against predictive accuracy.
- Analysed breakage-induced shifts in stiffness scaling, pressure exponents, and separability using paired no-breakage and particle-breakage DEM ensembles.
- Computed contact-fabric, normal-force, tangential-force, force-support, and cross-tensor anisotropy descriptors for contact-law-dependent micromechanical signatures.
- Produced publication-quality plots, parameter tables, validation diagnostics, and reproducible analysis scripts for manuscript preparation.

## Technical Skills

- **Programming:** Python, MATLAB, C; shell scripting; Fortran for Abaqus UMAT workflows
- **DEM & Sim:** YADE, LAMMPS; triaxial DEM; Hertz–Mindlin & Cundall–Strack; rigid-wall BCs; isotropic and deviatoric loading; contact-force post-processing
- **FEM / CAD:** Abaqus, ANSYS, SOLIDWORKS, CATIA V5, AutoCAD (supporting engineering tools)
- **Sci. Comp.:** NumPy, Pandas, SciPy, scikit-learn, Matplotlib; nonlinear regression; trust-region least squares; grouped K-fold CV; bootstrap; permutation tests; AIC/BIC; residual diagnostics
- **Mach. Learn.:** Neural networks, MLPs, regression, feature-sweep analysis, surrogate modelling, physics-aware machine learning
- **Mechanics:** Small-strain elasticity ( $E$ ,  $G$ ,  $K$ ,  $\nu$ ), contact mechanics, granular micromechanics, constitutive modelling, effective-medium benchmarks, wave propagation, fatigue & fracture
- **Data / Viz:** Publication-quality plotting; 2D/3D response surfaces; parity & residual plots; model-ranking tables; ParaView & VTK
- **Tooling:**  $\LaTeX$ , Git/GitHub, VS Code, Overleaf; technical manuscript writing; reproducible research workflows

## Computational and Data Analysis Methods

- Triaxial DEM simulation pipelines, audited elastic-window detection, and tangent-modulus extraction with elastic-consistency checks.
- Nonlinear least-squares calibration with trust-region optimization for separable stiffness laws.
- Model selection via AIC and BIC, grouped K-fold cross-validation, leave-one-level-out diagnostics, and bootstrap stability filtering.

- ▶ Physics-aware admissibility constraints: positivity, monotonic stiffening with confining stress, monotonic softening with void ratio, admissible Poisson's ratio.
- ▶ Surrogate modelling using neural networks and MLPs for cross-regime stiffness prediction.
- ▶ Computation of fabric and force-network anisotropy invariants (contact-fabric, normal-force, tangential-force, force-support, cross-tensor).
- ▶ Reproducible scripting workflows for DEM data ingestion, post-processing, regression, and figure generation.

### Selected Research Contributions

- ▶ Developed physics-aware, support-limited separable stiffness laws for DEM-derived small-strain moduli across confining stress, void ratio, and particle-modulus regimes.
- ▶ Quantified contact-law-dependent differences between Hertz–Mindlin and Cundall–Strack DEM packings in stiffness, Poisson's ratio, and anisotropy descriptor families.
- ▶ Integrated empirical constitutive modelling with neural-network and MLP-based prediction to compare interpretability against predictive accuracy.
- ▶ Built and audited elastic-window extraction and modulus-calculation workflows, including consistency checks across  $E$ ,  $G$ ,  $K$ , and  $\nu$ .
- ▶ Applied rigorous model selection and validation: AIC, BIC, grouped K-fold cross-validation, bootstrap filters, permutation tests, monotonicity / positivity admissibility.
- ▶ Extended stiffness modelling to breakage-aware DEM ensembles; quantified breakage-induced shifts in pressure exponents and separability.
- ▶ Connected macroscopic stiffness trends to microstructural descriptors (coordination number, fabric tensors, normal- and tangential-force anisotropy).

### Conference Presentations

- ▶ **Powders & Grains 2025.** Presenter, “Prediction of Small-Strain Properties of Dry Sand Using Curve Fitting and Machine Learning Models,” published in *EPJ Web of Conferences*, vol. 340, Article 09013, 2025.
- ▶ **ASME SSDM 2025.** Presenter, “Constitutive Modelling of Granular Materials Using Cundall–Strack and Hertz–Mindlin Contact Models,” Houston, TX, USA, May 5–7, 2025; published as Paper SSDM2025-152166 / V001T03A016.

### Selected Earlier Academic and Engineering Experience

Aug 2016 – Jul 2019 | ICFAI University, Raipur — Faculty of Engineering and Technology  
**Faculty Member**

- ▶ Three-year faculty appointment in the Faculty of Engineering and Technology; delivered undergraduate engineering courses and supervised B.Tech. projects and laboratory sessions.
- ▶ Contributed to curriculum development, examination and evaluation, lab coordination, and departmental academic activities; mentored undergraduate students on design, simulation, and project work.

2015 – 2016 | NIT Raipur / Tata Steel R&D, Jamshedpur  
**M.Tech. Thesis Research — API X65 Line Pipe Steel**

- ▶ Mechanical and microstructural characterization of API X65 line-pipe steel: tensile properties, fracture behaviour, Charpy impact, and high-cycle fatigue.
- ▶ Optical microscopy, hardness testing, fatigue / tensile systems, sample preparation, fractography, and Abaqus-based analysis; connected characterization to fracture and fatigue response relevant to structural integrity of line-pipe steel.

Jun – Jul 2012 | Korba, Chhattisgarh, India  
**Industrial Training — Hasdeo Thermal Power Station**

- ▶ Training in the boiler maintenance department; studied process flow from coal-based power generation through plant-efficiency considerations.
- ▶ Exposure to operations and maintenance in a thermal power generation environment.

May – Jun 2011 | Bhilai, Chhattisgarh, India  
**Industrial Training — Bhilai Steel Plant**

- ▶ Observed billet and bloom mill operations in the Steel Melting Shop.
- ▶ Studied process flow from ingot conversion to billet and bloom production.

## Projects

- 1. DEM-Based Small-Strain Stiffness Extraction Pipeline.** YADE-based triaxial simulation framework with audited elastic-window detection and consistent extraction of  $E$ ,  $G$ , and  $\nu$ .
- 2. Contact-Law Comparison: Hertz–Mindlin vs Cundall–Strack Packings.** Side-by-side DEM study quantifying contact-law effects on small-strain stiffness, coordination, and anisotropy.
- 3. Physics-Aware Separable Constitutive Modelling of Granular Elastic Moduli.** Calibration of separable stress–void-ratio–particle-modulus stiffness maps with admissibility constraints.
- 4. Breakage-Aware Stiffness Scaling and Machine-Learning Surrogate Modelling.** DEM ensembles with and without particle breakage; MLP surrogates for cross-regime stiffness prediction.
- 5. Fabric and Force-Anisotropy Descriptor Analysis for DEM Sphere Packings.** Descriptor-family partitioning of anisotropy invariants with bootstrap and permutation validation.

*Earlier engineering projects:* CAD modelling and analysis of radial-engine components in CATIA V5; multilayer cylinder stress analysis using SOLIDWORKS and ANSYS; structural analysis of a beam using SOLIDWORKS and ANSYS.

## Certifications and Training

- ▶ Computational Engineering & Simulations using Python: FEM & FVM, instructor-led online certificate course by iHUB DivyaSampark, IIT Roorkee and RBPL (Jun – Aug 2026).
- ▶ AutoCAD Certificate Course, NIT Raipur, Grade A (Aug – Sept 2011).
- ▶ C Language Certificate Course, Simply C, Grade A (May – Jun 2013).
- ▶ Workshop on MEMS, CAD Tools and Smart Structures (Jan 2015).
- ▶ Business Skill Development Programme (Mar 2012).

## Awards / Fellowships / Scholarships

- ▶ Branch Topper, B.E. Mechanical Engineering, CCEM, Raipur.
- ▶ First-year Topper, Mechanical Engineering branch.

## References

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